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Research Article
doi.org/10.1002/ejoc.202200623

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Expeditious Synthesis of a Glycopeptide Library

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Short proteoglycan fragments are of great importance for biochemical research. The solid-phase synthesis of such glycopeptides relies on excessive use of glycosylated amino acids, extended reaction times, and additional post-assembly deprotection protocols. We employed high-shear mixing for expedient and equimolar O-glycopeptide assembly. We further developed a stirring-based deprotection on the solid support, thus completing the synthesis of a glycopeptide library in a minimal amount of time and purification hurdles.

Introduction

Proteoglycans (PGs), a class of glycosylated proteins, are essential extracellular matrix components.[1,2] PGs are crucial for cellular function and recognition,[3–5] and are associated with various immunologic and neurologic diseases.[6–14]

The general structure of PGs consists of a core trisaccharide that connects the protein to glycosaminoglycan (GAG) chains. The xylose on the reducing end of the proteoglycan core glycan is bound to the protein through a glycosidic linkage to serine. The xylose is connected to two galactose units, with further linkage to the GAG by a uronic acid.[1] While extensive research is dedicated to the GAG moiety,[15] the effect of glycan and amino acid composition around the core glycosylation site on the biological function of PGs is somewhat neglected. One example of a noteworthy PG is Aggrecan, which is vital for the functioning of articular cartilage and the brain.[16,17] Aggrecan is estimated to contain up to 100 glycosylation sites, making the characterization of their linkage regions especially challenging.[18]

The characterization of such a vast variety of glycosylation sites requires the use of a large O-glycopeptide (GP) library.[19] Peptides and peptide libraries are prepared using the solid-phase peptide synthesis (SPPS) approach. However, the SPPS of GP libraries is hampered by difficulties associated with the glycosylated fraction.[20] The two main GP synthesis strategies are either to use a pre-synthesized glycosylated amino acid (GAA) in SPPS or to glycosylate a peptide after the assembly.[21] The latter strategy requires the use of special protecting groups, linkers, and conditions that are suitable both for oligosaccharide- and peptide-related reactions and shall not be discussed here. Using GAAs for GP synthesis is the most popular strategy, as it allows the installation of the glycan moiety in a precise location on the peptide by iterative SPPS steps. GP synthesis can be combined with chemoenzymatic approaches, but such processes are very sensitive to reaction conditions and have several limitations.[22,23] Other solid-phase methods reported recently also require a large excess of GAAs and are very time-consuming.[24,25]

Using pre-synthesized GAAs has several setbacks (Figure 1A). The preparation of GAAs is time-consuming, low-yielding, and very demanding synthetically.[26] As a result, the commercial availability of most GAAs is low, and the ones available are remarkably expensive. Second, while attempts at reagent recycling have been reported,[27] standard SPPS protocols rely on the use of high molar excess of amino acids to ensure reaction completion.[28] This sets a significant limit in GP synthesis considering the inaccessibility of GAAs. Third, GP synthesis is prone to side reactions, such as epimerization of the GAA during the synthesis or incomplete couplings due to the steric hindrance caused by the bulky GAA.[29] Finally, post-assembly processes are usually done in solution after cleaving the peptide from the support, which requires an additional purification step, leads to loss of material, and is time-consuming.[30,31] Due to these impediments, the synthesis of GPs is slow and demands large amounts of GAAs, and hence the synthesis of libraries is hardly achievable.

Accelerated synthesis is useful for providing access to a large number of peptides in a short time. By accessing peptide libraries in a short time,[32,33] extensive biological studies can be performed to determine structure-activity relationships.[34] Many coupling strategies were developed over the years to catalyze solid-phase reactions.[35] It was repeatedly demonstrated that new technologies that rely on a high temperature and improved mixing (e.g. flow or high-shear stirring) provide a route to accelerate SPPS processes.[34] We have recently reported High-Temperature Fast-Stirring Peptide Synthesis (HTFSPS), an accelerated and efficient method of SPPS.[33] A reactor was built which includes a heating jacket, overhead stirring, and a sintered glass filter. A feed insert double-acts as a baffle, which drastically improves the diffusion of reagents. The use of high-shear mixing at a high temperature allowed us to

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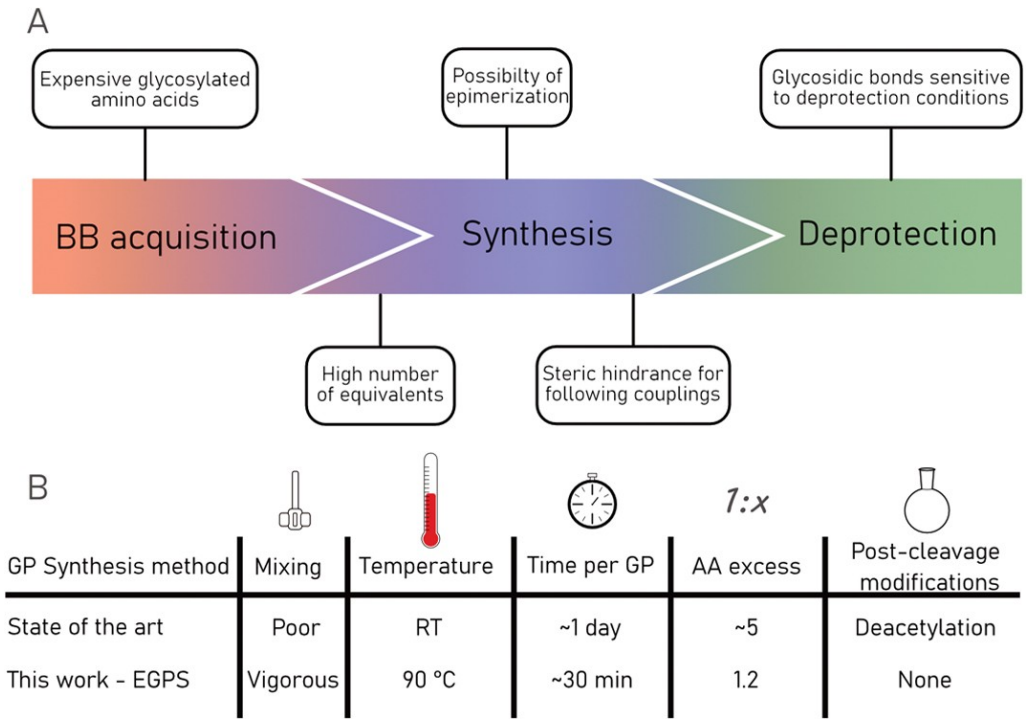
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B

GP Synthesis method	Mixing	Temperature	Time per GP	1:x AA excess	Post-cleavage modifications
State of the art	Poor	RT	~1 day	~5	Deacetylation
This work - EGPS	Vigorous	90 °C	~30 min	1.2	None

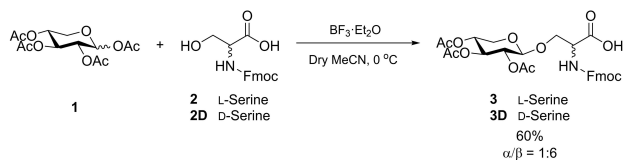
Figure 1. A) The challenges of GP synthesis; B) A comparison between the common methods in GP synthesis and this work.

synthesize peptides of diverse levels of complexity in minutes with minimal waste of reagents. Accelerating the synthesis of GP libraries will be an extremely valuable tool in determining the effect of sequence modifications on biological activity. Here, we describe the expeditious GP synthesis approach, which we term EGPS.

We have used HTFSPS methods to facilitate the synthesis of a complete GP of nine amino acids within 30 minutes, while using only 1.2 equiv. of GAA. We employed a fast on-resin deacetylation method that afforded a high-purity deprotected GP, thus minimizing post-assembly efforts. We then applied this expeditious protocol for the synthesis of an aggrecan-derived GP library within a fraction of the time such an endeavor would take using standard procedures.

Results and Discussion

We first aimed to synthesize a large amount of a model GAA (Scheme 1). The D-Xylopyranosyl tetraacetate glycosyl donor **1**[36] was used to glycosylate Fmoc-L-serine **2**[37] using boron trifluoride etherate to give Fmoc-protected xylosylated serine **3**[38–40] in 60% yield ($\alpha/\beta = 1:6$) on a gram scale.[41] Similarly, the D-Serine analog was prepared (SI). To develop the method for expeditious GP synthesis, we prepared a model peptide sequence LIS[Xyl(OAc)$_3$]GG (**TS1**, Figure 2) *via* different synthetic routes, utilizing HTFSPS principles but at different reaction



Scheme 1. Synthesis of xylosylated serine.

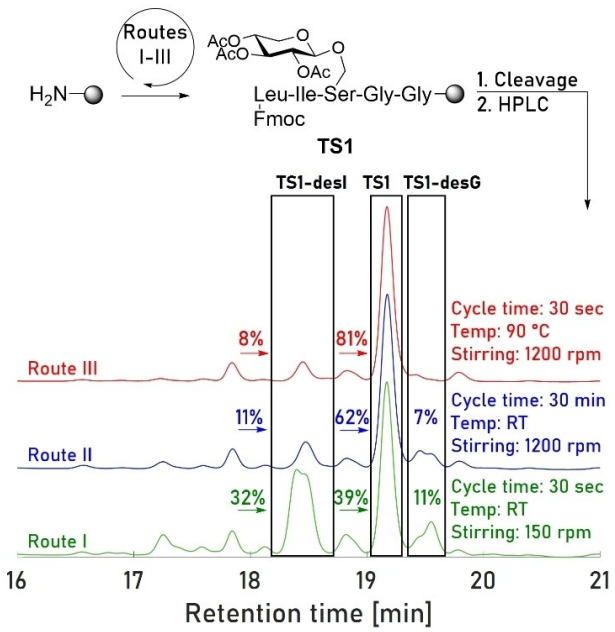


Figure 2. TS1 syntheses *via* routes I–III.

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conditions (Routes I–III, Table 1 and Figure 2, see experimental for details). Two equiv. of amino acids, including GAA **3**, were used for each coupling, while the coupling time, temperature, and stirring rates were changed. After each synthesis, **TS1** was cleaved and its purity was determined by HPLC.

Our first attempt at the synthesis of **TS1** was done following route I, which employs a short reaction time at room temperature with mild stirring (Table 1). **TS1** was obtained at 39% purity, while deletions of isoleucine, **TS1-desI** (32%), and of glycine, **TS1-desG** (11%), were also observed (Figure 2, green).

The presence of **TS1-desI** implies that the coupling of Ile to the GAA is difficult due to steric hindrance by the bulky glycosylated serine. The presence of **TS1-desG** shows that the coupling of the glycine directly to the rink amide linker is also not trivial under these conditions.

Our next attempt was done following route II, which employs longer reaction times, a higher stirring rate, and is carried out at RT. Using route II conditions, **TS1** was obtained at a crude purity of 62%, exhibiting only 11% of **TS1-desI** and 7% of **TS1-desG** (Figure 2, blue).

This indicated that a high stirring rate and long reaction times decrease deletions during synthesis. We attempted to shorten the coupling time by synthesizing **TS1** *via* route III, which employs only 30 seconds per coupling while maintaining a constant temperature of 90 °C and fast stirring at 1200 rounds per minute (rpm). *Via* route III, **TS1** was synthesized at a crude purity of 81%, with **TS1-desI** diminished to 8% and **TS1-desG** almost completely vanishing (Figure 2, red). These results indicate that an accelerated synthesis of GPs is possible using HTFSPS strategy. It also implies that the combination of a high temperature and fast stirring enables GP synthesis using much lower amounts of GAA than a standard SPPS process even when short coupling and deprotection cycles are employed. We concluded that route III conditions could be very useful for accelerated GP synthesis.

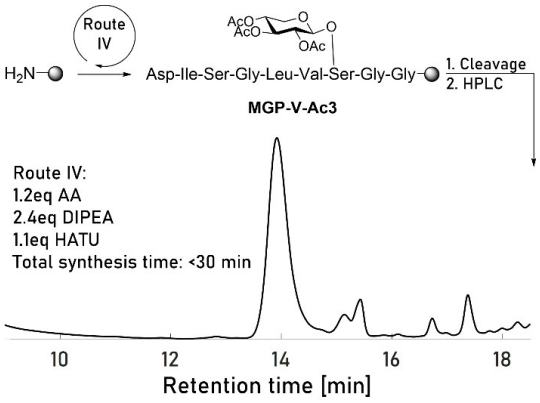
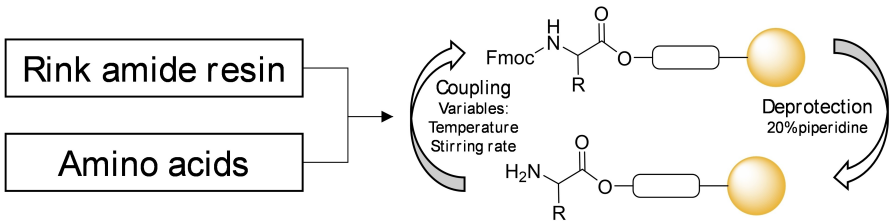


Figure 3. HPLC chromatogram of crude **MGP-V-Ac3** synthesized *via* route IV.

Racemization of GAAs has been previously reported, and the synthesis of peptides under a high temperature is known to increase the formation of undesired epimers.[29] To evaluate the possibility of racemization, the D analog of **TS1**, **TS1-D**, was synthesized using xylosylated D-serine **3D** (Scheme 1 and SI). Comparative HPLC analysis showed that the synthesis of **TS1** using route III conditions did not yield traceable amounts of **TS1-D**, proving that the process does not result in significant GAA epimerization.

After establishing the conditions for accelerated synthesis, we applied them to prepare the peptide fragment DISGLVS[Xyl(OAc)$_3$]GG (**MGP-V-Ac3**, Figure 3), which is similar to one commonly found in aggrecan.[18] It was synthesized *via* route III and obtained at a crude purity level of 80% (SI). As this synthesis was successful, we have devised synthetic route IV

Table 1. Conditions of synthetic routes used in this work.



Route number[a]	Temperature [°C]	Coupling time	Stirring rate [rpm]	Deprotection time[b]	Reactant equivalents
I	RT	30 sec	150	30 sec	2 eq AA 4 eq DIPEA 1.9 eq HATU
II	RT	30 min	1200	2x5 min	2 eq AA 4 eq DIPEA 1.9eq HATU
III	90	30 sec	1200	30 sec	2 eq AA 4 eq DIPEA 1.9 eq HATU
IV	90	30 sec	1200	30 sec	1.2 eq AA 2.4 eq DIPEA 1.1 eq HATU

[a] All reactions were performed in 3 mL of DMF in the same overhead-stirring reactor on 100 mg of rink amide resin (loading 0.48 mmol/g). [b] Fmoc deprotection was performed in 3 mL of 20% v/v piperidine solution in DMF.

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(Table 1), which differs from route III only by using 1.2 equiv. of amino acids, including GAA **3**, instead of 2 as used in previous protocols. The synthesis of **MGP-V-Ac3** was carried out *via* route IV, and the product was collected at a high crude purity (Figure 3).

This is a clear indication of the strategy efficiency, which allows the use of practically equimolar amounts of AAs where previously a large excess was considered necessary. This result suggests that the synthesis of GPs could be achieved with a minimal amount of GAAs.

In a standard SPPS protocol, the protecting groups from the glycans must be deacetylated from the assembled GP after its cleavage from the resin to provide a biologically relevant entity. Following this strategy, **MGP-V-Ac3** was cleaved from the solid support, purified by preparative HPLC, lyophilized, treated with a solution of 1 M NaOMe in MeOH for 90 min, then quenched with AcOH, and purified again by preparative HPLC to provide **MGP-V** (Figure 4, method D1, green). This strategy adds two post-cleavage steps, which take a significant amount of time compared to the synthesis itself and may lead to additional side reactions and loss of product. To avoid these two post-cleavage steps and their potential disadvantages, we wished to conduct the deacetylation reaction on the resin. On-resin deacetylation was previously reported but takes more than an hour to perform.[41] We assume this extended reaction time results from the deficient diffusion of the reagents into the support during the Zemplén reaction, in which the methanol used as a solvent shrinks the polystyrene beads and limits permeation of the base to the anchored protected GP.[42] As the fast stirring used in HTFSPS provides very efficient diffusion to and into the beads, we hypothesized that it would facilitate deacetylation on the solid support even when MeOH is used as a solvent. We attempted two new procedures for the deacetylation of solid-phase-anchored **MGP-V-Ac3** while using very fast stirring. The deacetylation was done at room temperature under constant stirring of 1200 rpm by treating the resin with 3 mL of a solution of 160 mM NaOMe in MeOH either twice for 10 minutes (method D2) or thrice for 5 min (method D3).

Deacetylation using method D2 (Figure 4, blue) provided the fully deprotected GP **MGP-V** in 74% crude purity, which proved that fast stirring can be used for deacetylation on the solid support.

However, partially deacetylated **MGP-V-Ac** and the fully acetylated **MGP-V-Ac3** were still present after using method D2 protocol, proving that the conditions are not optimal.

We tested a second deacetylation protocol on the solid support, method D3, which utilized three cycles of deprotection, albeit at a shorter time. Applying method D3 (Figure 4, red) increased the purity of **MGP-V** to 92% and the partially-acetylated products almost disappeared. An attempt to deacetylate **MGP-V-Ac3** using hydrazine hydrate solution[43] proved less efficient than the Zemplén conditions (See SI).

Both of these methods present a considerable improvement compared to the in-solution method D1, both in terms of reaction time and in terms of avoiding additional work-up and purification stages. They also exhibit a substantial decrease in reaction times compared to previously reported solid-phase protocols. We infer this indicates that our stirring was effective enough in improving the diffusion to allow the deacetylation reaction to occur quickly on the solid support, despite the shrinking effect of the solvent. Interestingly, we found that a third repeat of the reaction improves its efficiency, even if the

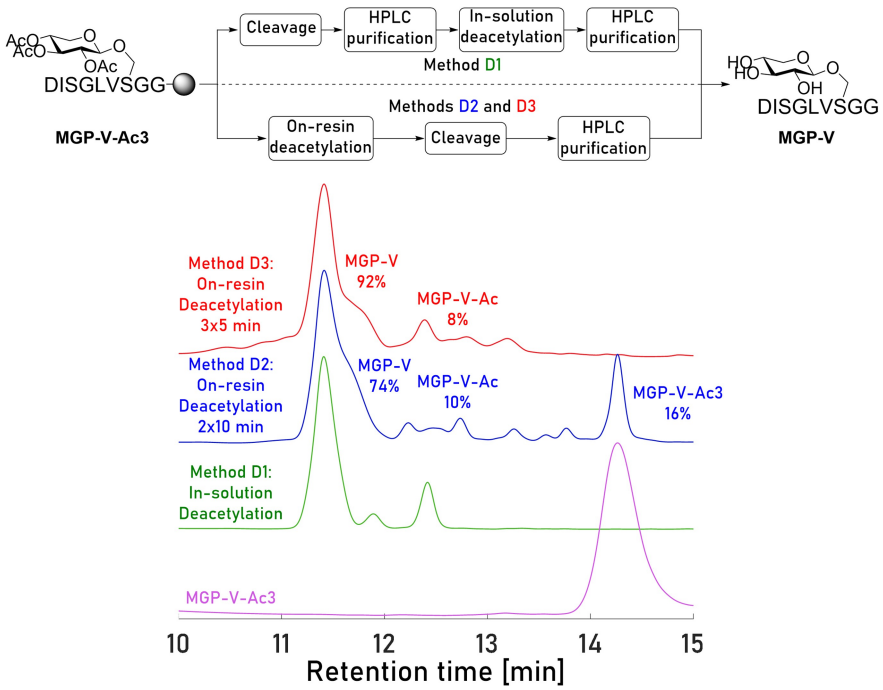


Figure 4. Top: Pathways of deacetylation: In solution (method D1) and on-resin (methods D2 and D3). Bottom: Chromatograms of crude products of deacetylation *via* methods D1-D3.

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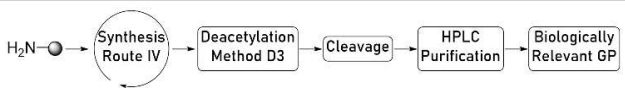
total reaction time is decreased. We believe this difference stems from various side reactions, such as hydrolysis, which quench the nucleophile after a relatively short time, leading to a fast decline in the reaction rate. Consequently, the addition of a fresh portion of NaOMe solution instead of the quenched one gives the kinetics of the reaction a significant boost. The results indicate this decline in reaction rate happens well within the timeframe of 10 min, and thus method D3 provides better efficiency than method D2 in a shorter time.

Following this improvement, we have named the synthetic procedure which combines route IV with deacetylation method D3 as EGPS – **E**xpeditious **G**lyco**p**eptide **S**ynthesis and used it to prepare a library of GPs with the nine most common XSG sequences in aggrecan.[18] We have systematically changed the AA introduced after the xylosylated serine **3** (marked red in Table 2), as it previously proved to be a difficult coupling. Additionally, To test EGPS on a completely different GP, we have synthesized the sequence ISAS[Xyl]GVE, which is another common motif in aggrecan (Table 2).

All GPs were obtained at a sufficient amount and a relatively high degree of crude purity. As some of the variable AAs contain bulky groups, both hydrophobic and hydrophilic, this serves as a good indication of the potential of EGPS.

Additionally, EGPS was successfully used to synthesize a completely different sequence from the bulk of the library, implying that its use is not limited to select sequences. All GPs were purified to a high degree within one HPLC step, thus minimizing the loss of material in the process compared to a standard post-cleavage strategy. All peptides were characterized by HPLC, MS, and ^{1}H-NMR. Each GP required only about 34 mg of GAA, and complete synthesis of a deprotected GP was done in less than an hour. In comparison, a state-of-the-art peptide synthesizer would require about four hours to synthesize a GP of the same length, without deprotection, which takes another 12 h, and 142 mg of GAA would be used in the process.[24] We proved that 10 GPs could be synthesized with less than 350 mg of GAA, and that the assembly of such a library could be done in one or two working days, compared to weeks of work and roughly 1.5 g of GAA in the standard protocol. Such a quick and efficient synthesis of a GP library has not been reported so far and could have great potential for biochemical research. As this process is still under development, these results are likely to be achievable for many other GPs in the future.

Table 2. The synthesized Aggrecan-derived GP library.



Sequence	Calculated mass (*m/z*)	Observed mass (*m/z*)	Crude purity[a]
DISGL**V**S[Xyl]GG	934.461	935.4685 $[M+H]^+$	92%
DISGL**P**S[Xyl]GG	932.445	955.512 $[M+Na]^+$	83%
DISGL**A**S[Xyl]GG	906.429	929.520 $[M+Na]^+$	82%
DISGLIS[Xyl]GG	948.476	971.607 $[M+Na]^+$	69%
DISGLLS[Xyl]GG	948.476	971.565 $[M+Na]^+$	34%
DISGL**G**S[Xyl]GG	892.414	893.4216 $[M+H]^+$	79%
DISGL**S**S[Xyl]GG	922.424	923.4362 $[M+H]^+$	77%
DISGL**T**S[Xyl]GG	936.440	959.608 $[M+Na]^+$	55%
DISGL**Y**S[Xyl]GG	998.456	1021.600 $[M+Na]^+$	63%
ISAS[Xyl]GVE	792.387	815.498 $[M+Na]^+$	64%

[a] Crude purity was calculated by absorbance at 220 nm.

Conclusion

We presented EGPS, an accelerated approach for the synthesis of GPs. We have shown that synthetic GP-related hurdles like difficult couplings and inefficient glycan deprotection could be overcome by applying high-shear mixing. EGPS enabled the synthesis of GPs using an equimolar quantity of GAA in record time. EGPS synthesis of ten aggrecan-derived GPs was done in hours and with a tiny amount of GAA proving its value for peptide library preparation. This quick, economical, and easy-to-perform method could herald the possibility of synthesis of a wide array of hitherto nearly unobtainable GP libraries, which could be used for biological research. We believe that EGPS advantages could be utilized in the future for the synthesis of GPs with complex glycans and with multiple glycosylation sites.

Experimental Section

The synthetic method used for GP synthesis is similar to the one described previously for HTFSPS.[33] All syntheses were carried out in the same reactor, using 100 mg of Rink amide resin (loading 0.48 mmol/g). Couplings were done in one of four synthetic routes which differ by their conditions, described in Table 1. Amino acids and activators were dissolved in N,N-dimethylformamide (DMF) in a total volume of 3 mL (1 mL of each solution, concentrations in Routes I–III: 32 mM AA, 64 mM DIPEA, 30.4 mM HATU. Concentrations in Route IV: 19.2 mM AA, 38.4 mM DIPEA, 17.6 mM HATU). Fmoc deprotections were done in 3 mL of a 20% v/v solution of piperidine in DMF. After every stage (couplings and deprotections) a wash was done by stirring the resin for 30 seconds with 3 mL of DMF. After each stage, the solvent was filtered out and then the new reactants were injected into the reactor. When the peptide synthesis was complete, the reactor was allowed to cool down (if required) while washing the resin with DMF, then dichloromethane (DCM), then methanol, and eventually Et_2O. When On-resin deacetylation was performed, it was done after the wash with methanol, with a solution of 160 mM of NaOMe in methanol. After each repetition of the deacetylation reaction, the resin was washed for 30 seconds with 3 mL of methanol. Deacetylation with 10% v/v hydrazine hydrate in DMF was also attempted, but was found to be less efficient than the Zemplén procedure. After the resin was dried, cleavage was done with a solution of 5.52 mL trifluoroacetic acid (TFA), 0.3 mL of triple-distilled water (TDW), and 0.18 mL of triisopropylsilane, then the product was precipitated in a cold solution of diethyl ether and hexane (1:1) and lyophilized.

Acknowledgements

M.H. and D.B.A.A are supported by COST Action 18103 "INNOGLY" and European Innovation Council project "Smellodi" funded by the European Union.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Deacetylation · Glycopeptides · Glycosylated amino acids · Solid-phase synthesis · Synthetic methods

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Manuscript received: May 30, 2022
Revised manuscript received: July 4, 2022
Accepted manuscript online: July 5, 2022

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